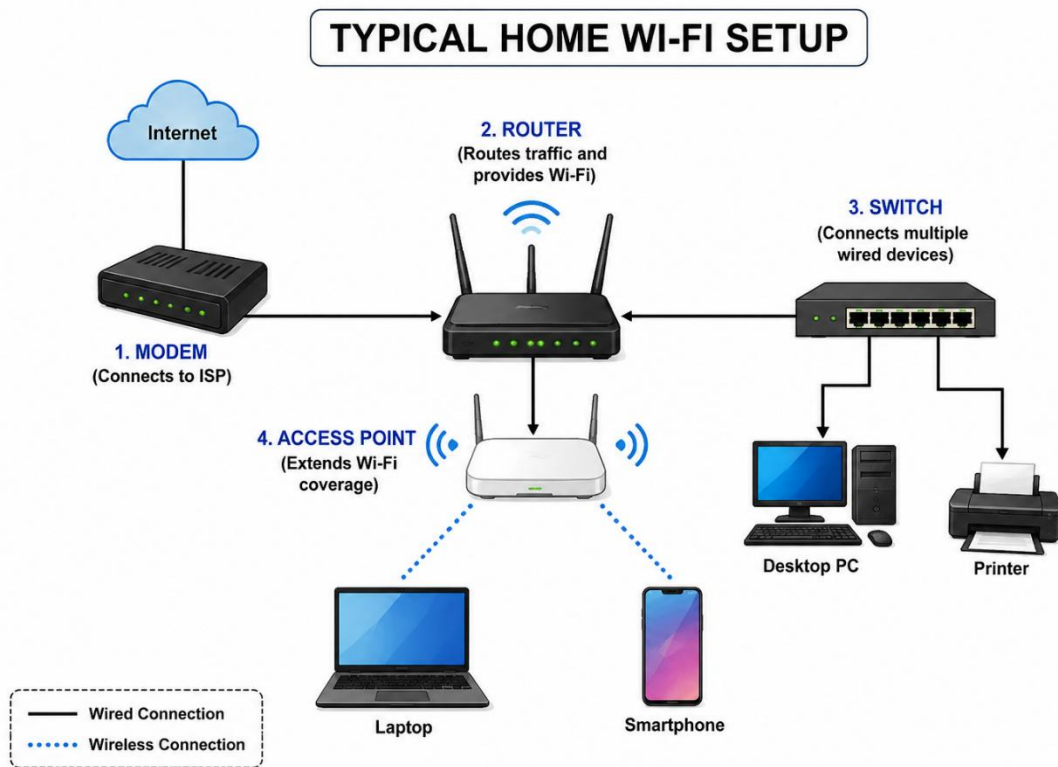


1. Typical Home Wi-Fi Setup Diagram



2. Networking Hardware Functions & Examples

Device	Function	Example
Modem	Connects your home network to the Internet Service Provider (ISP).	Airtel or Jio Fiber modem at home.
Router	Shares internet between multiple wired and wireless devices.	Home Wi-Fi router used for YouTube, Netflix, and browsing.
Switch	Connects multiple wired devices in a Local Area Network (LAN).	Used in offices to connect many computers.
Access Point	Extends or provides wireless (Wi-Fi) connectivity.	Installed in schools, malls, hotels, and colleges.
Network Cable (Ethernet)	Transfers data between networking devices using a wired connection.	Connecting a desktop PC to a router for faster internet.

3: Types of Network Cables



Ethernet (Cat5e/Cat6)



Coaxial Cable



Fiber Optic Cable

3. Types of Network Cables

- **Ethernet Cable:** Commonly used for connecting computers, routers, and switches within a Local Area Network (LAN).
- **Coaxial Cable:** Commonly used for connecting cable television (CATV) and providing broadband internet from the wall to your cable modem.
- **Fiber Optic Cable:** Commonly used by ISPs to deliver high-speed, long-distance internet into homes and businesses.

4: Comparison of Router & Switch

Feature	Router	Switch
Purpose	Connects different networks (LAN to Internet).	Connects devices within the same LAN.
Data Handling	Uses IP addresses to forward packets.	Uses MAC addresses to forward frames.
Location	Home, office, or ISP connection.	Inside offices, schools, and computer labs.
Internet	Yes, shares internet connection.	No, only connects local devices.
Security Features	Can provide firewall, NAT, and DHCP.	Usually does not provide firewall or NAT.
No. of Networks	Connects multiple networks.	Works within a single network (LAN).
Wireless Support	Usually provides Wi-Fi.	Normally does not provide Wi-Fi.